NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

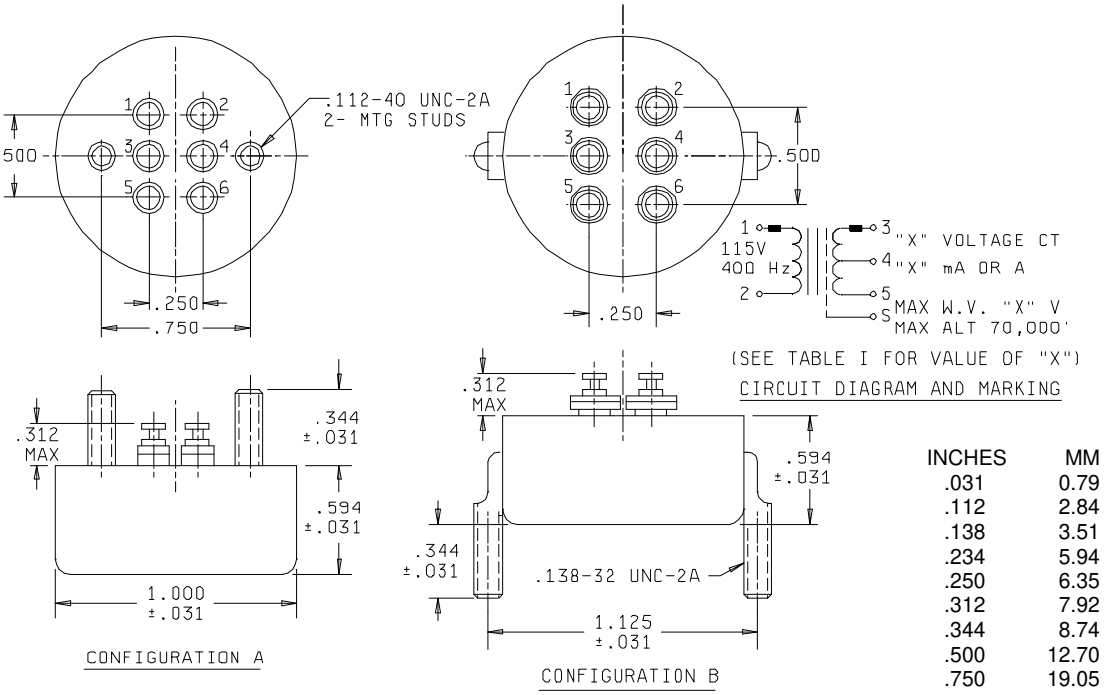
MIL-PRF-27/51C
7 February 1984
SUPERSEDING
MIL-T-27/51B
12 May 1978

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 3 VOLT AMPERES, 400 HZ, ELECTROSTATIC SHIELDED, TF4S03YY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the transformers described herein shall consist of this specification and the latest issue of MIL-T-27.



INCHES	MM
.031	0.79
.112	2.84
.138	3.51
.234	5.94
.250	6.35
.312	7.92
.344	8.74
.500	12.70
.750	19.05
.812	20.62
1.000	25.40
1.125	28.58

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.015 (0.38 mm).
4. Marking shall be on the side of the case.
5. Terminals are color coded for permanent identification in addition to terminal marking shown. Ground terminal "S" is black.
6. Secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configuration.

Ⓒ denotes changes

FSC 5950

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:19Z
Check the source to verify that this is the current version before use.

REQUIREMENTS: (When numbers in parentheses i.e. (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Primary voltage (1-2): 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 1.7 ounces, maximum.

Duty cycle: Continuous.

Case: Metal enclosed.

Material: Steel.

Altitude: 70,000 feet, maximum.

Operating temperature range: -55°C to +130°C maximum.

Terminals: Feed thru.

Terminal height: 0.312 inch, maximum.

Terminal strength: Method 211 of MIL-STD-202, test condition A, 5 pounds.

Dielectric withstanding voltage:

At sea level: 2.8 times the maximum working voltage specified in table I.

At reduced barometric pressure: 1.25 times the maximum working voltage specified in table I.

Electrical characteristics:

No load: With 115 volts, 400 Hz across (1-2): Current in (1-2) shall not exceed 6.0 mA rms.

Rated load: With 115 volts, 400 Hz across (1-2) and a resistive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I.

Efficiency: 75 percent minimum at rated load.

Electrostatic shielding: The voltage ratio shall be a minimum of 5 to 1 at 20 kHz.

Polarity: Additive with terminals 2 and 3 connected.

Regulation: $\frac{\text{Voltage (no load)} - \text{Voltage (rated load)}}{\text{Voltage (rated load)}} \times 100$ shall not exceed 12 percent.

Phase shift: 1.0 electrical degree between primary and secondary voltage at rated load, maximum.

Temperature rise: 20°C, maximum with 115 volts, 400 Hz across (1-2) at an ambient temperature of 110°C.

Vibration: MIL-STD-202, method 204, test condition D.

Marking location: See figure 1.

© QUALITY ASSURANCE PROVISIONS:

Extent of qualification:

Source: https://assist.dla.mil -- Downloaded: 202610-16T03:19Z
Check the source to verify that this is the current version before use.

Qualification testing and approval to M27/49-44 A or B, M27/51-44 A or B, M27/53-44 A or B, M27/56-44 and M27/66-27 shall be sufficient to grant qualification approval to MIL-T-27/48 through MIL-T-27/66 inclusive, all parts.

Qualification testing and approval to M27/51-44 shall be sufficient to grant qualification approval to M51/39-01 through M27/51-44.

Part number example:

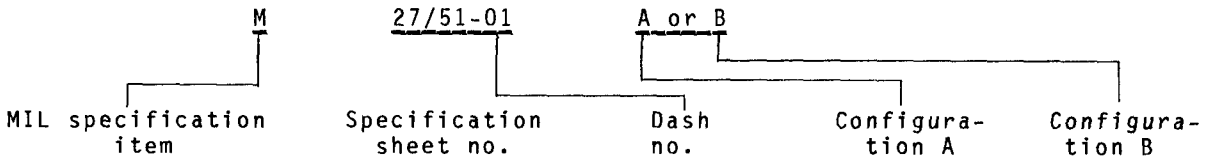


TABLE I. Electrical ratings.

Dash no.	Secondary voltage (3-5) (V rms) ±3%	Secondary current (3-5) (amps rms)	Working voltage (V rms max)
01 A or B	6	0.500	325
02 "	6.3	0.476	"
03 "	7	0.429	"
04 "	7.5	0.400	"
05 "	8	0.375	"
06 "	9	0.333	"
07 "	10	0.300	"
08 "	11	0.273	"
09 "	12	0.250	"
10 "	12.6	0.238	"
11 "	13	0.231	"
12 "	15	0.200	"
13 "	16	0.188	"
14 "	18	0.167	"
15 "	20	0.150	"
16 "	22	0.136	"
17 "	24	0.125	"
18 "	26	0.115	"
19 "	28	0.107	"
20 "	30	0.100	"
21 "	33	0.091	"
22 "	36	0.083	"
23 "	39	0.077	"
24 "	43	0.070	"
25 "	47	0.064	"
26 "	51	0.059	"
27 "	56	0.054	"
28 "	62	0.048	"
29 "	68	0.044	"
30 "	75	0.040	"
31 "	82	0.037	"
32 "	91	0.033	"
33 "	100	0.030	"
34 "	110	0.027	"
35 "	115	0.026	"
36 "	120	0.025	"
37 "	130	0.023	"
38 "	150	0.020	"
39 "	180	0.017	"
40 "	200	0.015	440
41 "	220	0.014	440
42 "	240	0.013	440
43 "	270	0.011	580
44 "	300	0.010	580

Source: https://assist.dla.mil -- Downloaded: 202640-16T03:19Z
Check the source to verify that this is the current version before use.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - MC, AS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5950-0623-17)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:19Z
Check the source to verify that this is the current version before use.